WARWICK
THE UNIVERSITY OF WARWICK

Lecture 7
Text Classification

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Natural Language Processing
CS918

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Outline

- What is Text Classification?
  - Examples of Text Classification
- Supervised Text Classification
  - Sentiment Analysis
- Evaluation
- Error Analysis

What is Text Classification?

Text
d

Class
c

finance

technology

politics

Training set

Learn a function to map from **d** to **c**

finance

Predicted Class
c*

What is Text Classification?

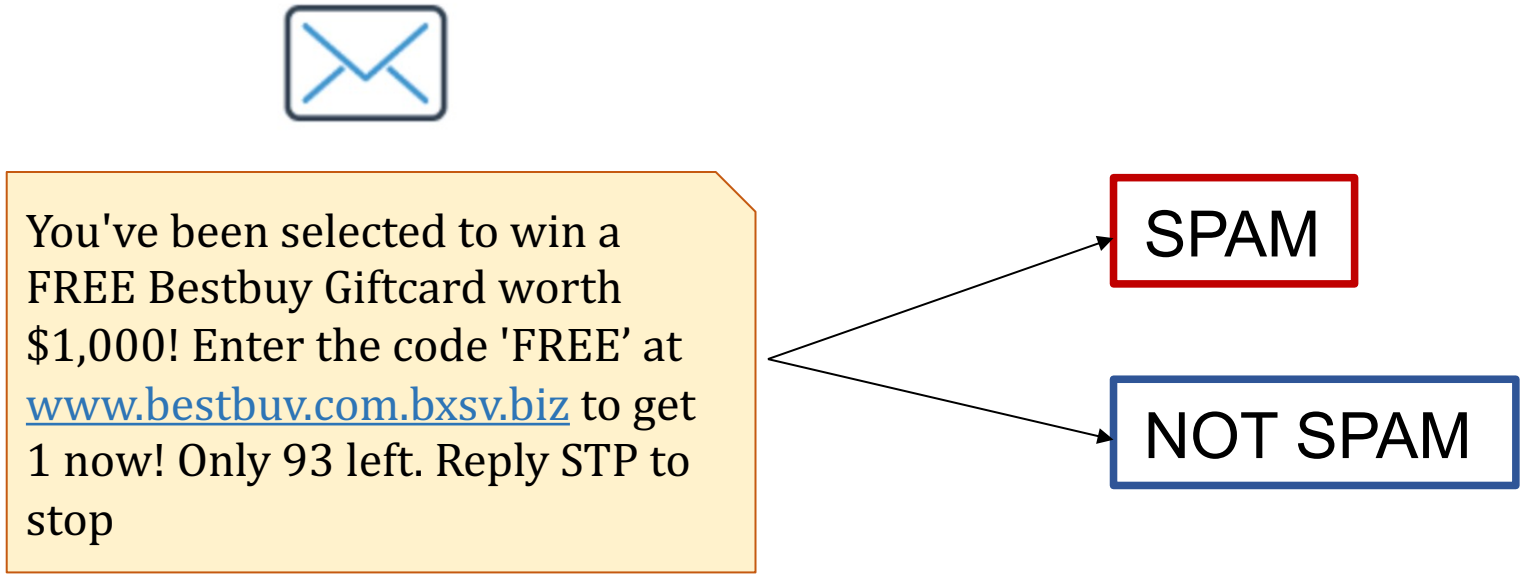
- Having as input:
  - A **text document d**
  - A **set of categories** $C=\{c_1, \ldots, c_m\}$

- The **text classification** task **outputs**:
  - Predicted class **c*** that **document d belongs to**.

$$c^* \in C$$

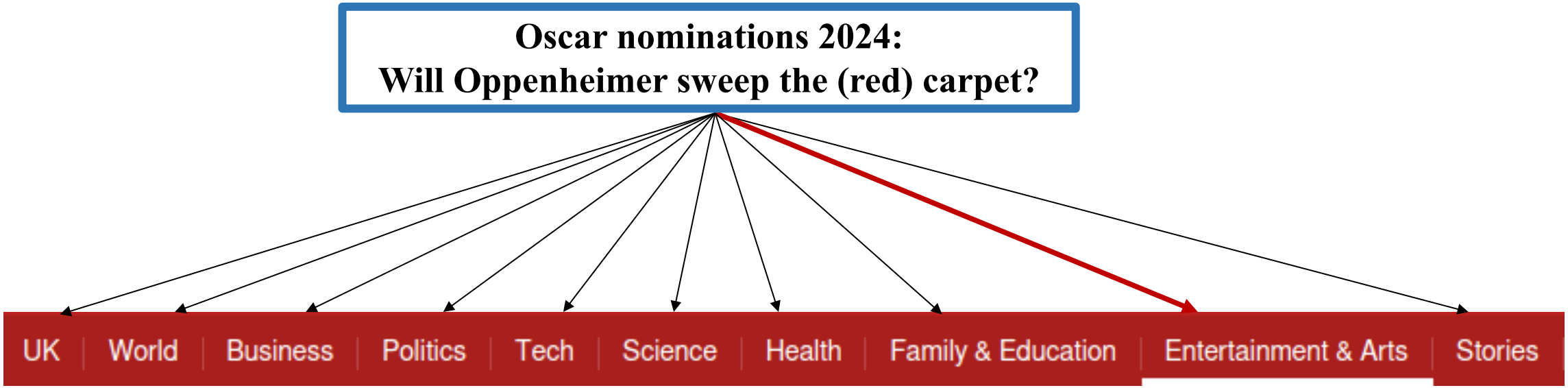
Examples of Text Classification

- **Spam detection**: classifying emails/web pages as spam (or not)



Examples of Text Classification

- **Classification by topic**: what is the text about?



Examples of Text Classification

Sentiment analysis: is a text positive, negative or neutral?

☺ I really **liked** the food at the restaurant.

😐 We were 8 friends who went there for the first time.

☹ The service was **terrible**.

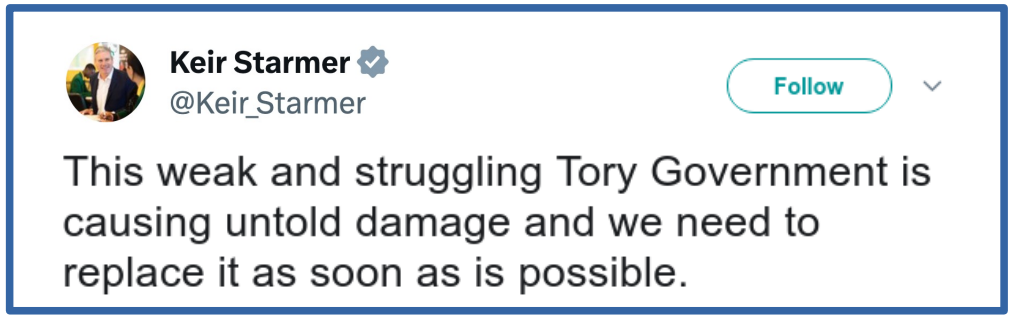
Examples of Text Classification

- **Language identification**: what language a text is written in?

Wieviel Uhr ist es? → {German, English, Spanish, French}

Examples of Text Classification

- **Classification of political orientation**:
  does a text support labour or conservative?

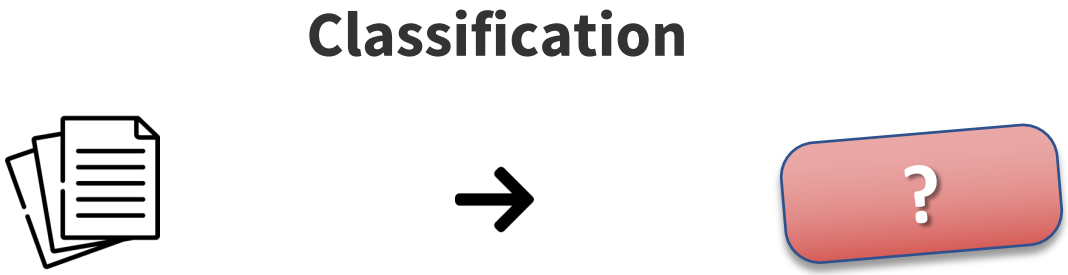


Labour

Conservative

What is Text Classification?

- A range of different problems, with a **common goal**:
  - Assigning a **category/class to each document**
  - We **know the set of categories** beforehand



What is Text Classification?

- A range of different problems, with a **common goal**:
  - Assigning a **category/class to each document**
  - We **know the set of categories** beforehand

Classification

→

Topic	{Business, Sports, … }
Sentiment	{Positive, Neutral, … }
Language	{English, Chinese, … }
Political Orientation	{Conservative, Labour, …}
… and many others!	

Text Classification: Approaches

Rule-based classifiers, e.g. if a sentence contains '*excellent*' → positive polarity

- Significant manual effort involved.
  - *Catch:* How many different ways to express the same positive feeling?

Supervised classification:

- **Given**: a hand-labeled set of **document-class pairs**

$$(d_1,c_1), (d_2, c_j), \ldots, (d_m,c_2) \rightarrow \text{classified into } C = \{c_1, \ldots, c_j\}$$

- The classifier **learns a model** that can **classify new documents into C**.

Supervised Text Classification

Supervised Text Classification

- **<u>Assumption</u>:** We have a manually labelled dataset, e.g.:

  $\mathbf{d_1}$: 'That's really good, I love it' → **positive**

  $\mathbf{d_2}$: 'It was boring, don't recommend it' → **negative**

  …

  $\mathbf{d_n}$: 'I wouldn't go again, awful' → **negative**

- **If not**, we need to **find one** or **label one ourselves**.

Supervised Classification: **Decisions to Make**

1. **Split the dataset** into train/dev/test sets.
   - or often just train/test.

2. What **features** are we going to use to represent the documents?

3. What **classifier** are we going to use?
   - Choose **settings, parameters**, etc. for the classifier.

1. Splitting the Dataset: The problem of Generalisation

- We can **split the dataset into 3 parts**:
  - Training set → the largest set as we want proper training
  - Development set
  - Test set

  - **Tweak classifier based on the development set**, then test it on the test set.
    - **Tweaking and testing on the test set may lead to overfitting** (doing the right things specifically for that test set, not necessarily generalisable)

What is Overfitting?

- **Overly adjusting** our classifier and features **to a specific test set**.

- The classifier **may not generalise** to new instances beyond the current test set.

Pretty much like designing a mattress based on a single observation of sleeping position!

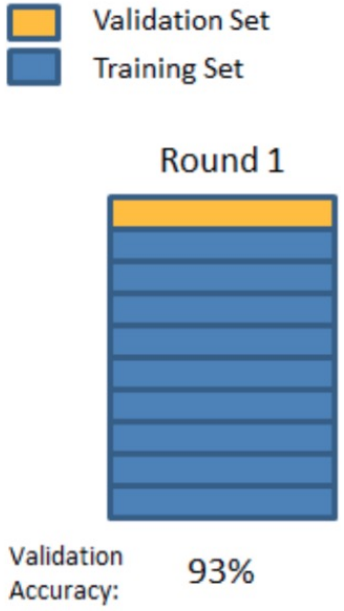
How to Avoid Overfitting?

- Having train/dev/test, **tweak our classifier for the dev set**.
  - Then apply to the **test set**, does it **generalise**?

- **Cross-validation**.

Cross-Validation

- **Cross-validation:** train and test on different “folds”
  - e.g.: 10-fold cross-validation, split the data into 10 parts.
  - each time 1 fold is used for testing, the other 9 for training.
  - after all 10 runs, compute the average performance.



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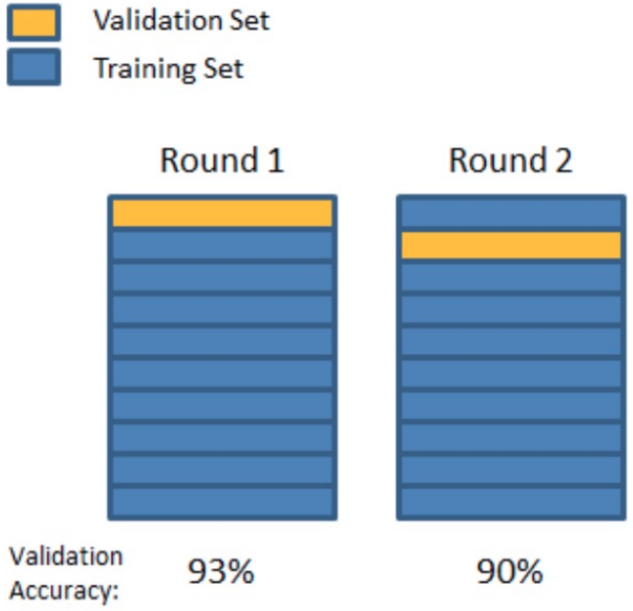

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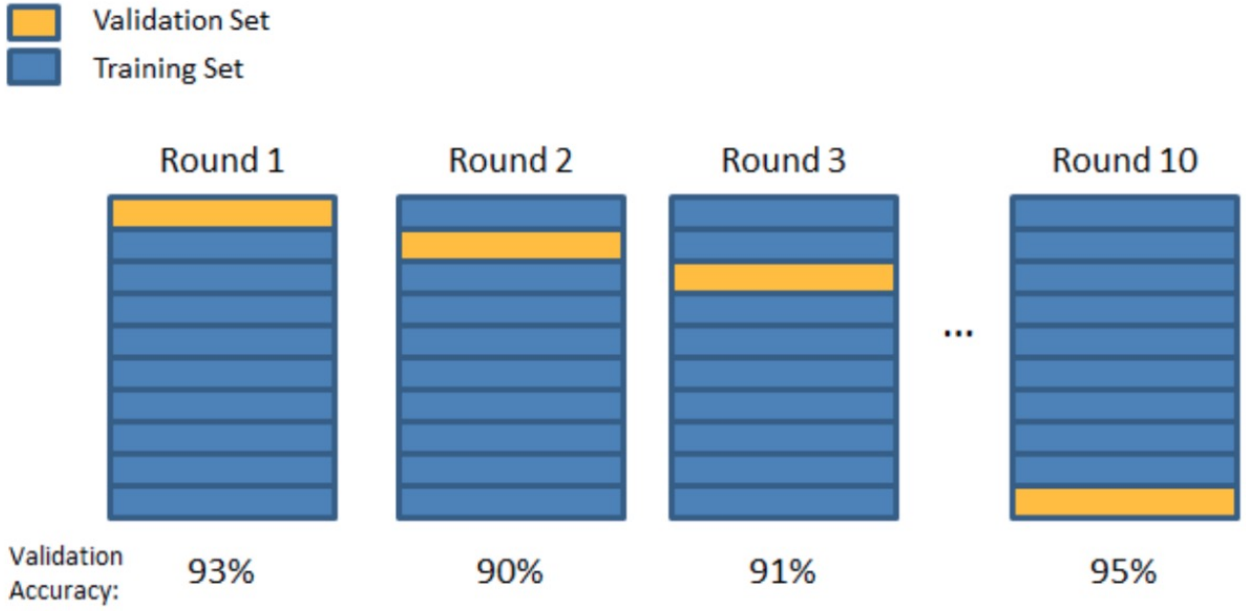


2. Choosing the Features

- Usually start with some basic features:
  - Bag of words.
  - Or preferably word embeddings.

- Keep adding new features:
  - Need to be creative.
    Think of features could characterise the problem at hand.

Thinking of Features

Possible features:

- **Sentiment analysis** → *counts* of positive/negative words.
- **Language identification** → probabilities of *characters* (how many k's, b's, v's…), features from word suffixes (e.g. many -ing words → English)
- **Spam detection** → count words in *blacklist*, domain of URLs in email (looking for malicious URLs)

Choosing the Features

- How to assess which features are good?

- Empirical evaluation:
  - Incremental testing:
    keep adding features, see if adding improves performance
  - Leave-one-out testing:
    test all features, and combinations of all features except one.
    when leaving feature *i* out performs better than all features, remove feature *i*
  - Error analysis: (later in this lecture)
    look at classifier's errors, what features can we use to improve?

3. Choosing a Classifier

- Many different classifiers exist, well-known classifiers include:
  - **Naive Bayes**
  - **Logistic Regression** (Maximum Entropy classifier)
  - **Support Vector Machines** (SVM)

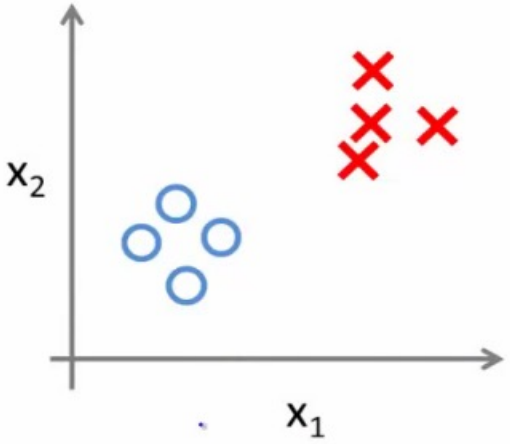
- Classifiers can be **binary** ($k = 2$) or **multiclass** ($k > 2$).

3. Choosing a Classifier

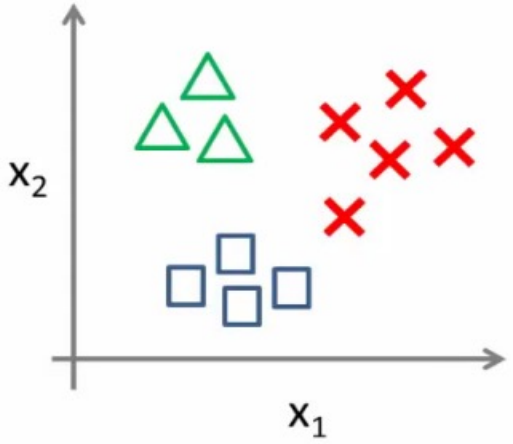
- How many categories (k)?
  - [k=2] **Binary** → binary classifier
  - [k>2] **Multiclass**:
    - One-vs-all classifiers
      Build *k* classifiers, each able to distinguish class *i* from the rest. Then combine output of all classifiers (e.g. based on their confidence scores)
    - Multinomial/multiclass classifiers

Binary vs. Multiclass Classifier

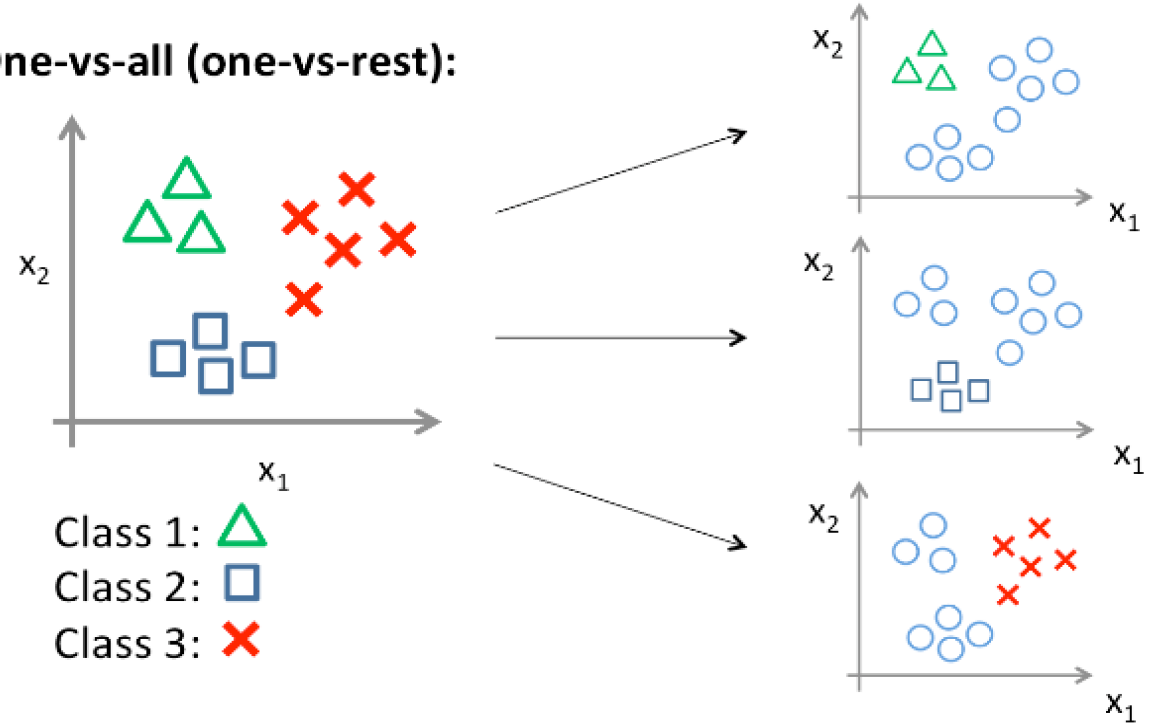
Binary classification:



Multi-class classification:



One-vs-all (one-vs-rest):



When to use Multinomial or One-vs-All?

- **Multinomial**:
  - is generally faster, a single classifier.
  - classes are mutually exclusive, no overlap.
- **One-vs-all**:
  - **Multilabel classification**, document can fall in **1+ categories**:
    e.g. classify by language:
    I said “bonjour mon ami” to my friend → English & French
    How? Out of *k* classifiers, those with **confidence > threshold**

Sentiment Classification

Sentiment Analysis

Is a text **positive**, **negative** or **neutral**?

☺ I really **liked** the food at the restaurant.

😐 We were 8 friends who went there for the first time.

☹ The service was **terrible**.

- It tastes like beer! ?
- It tastes like my mom said it would! ?
- If it was served with milk, it would taste great! ?

How to Build a Sentiment Classifier?

Two main approaches:

- **Lexicon-based**: Unsupervised classifier using lexicons
- **Learning-based**: Supervised classifier leveraging training data

Lexicon-Based Sentiment Classification

- Using **lexicons** of positive and negative words, e.g.:

```
fantastic
fantastically
fascinate
fascinating
fascinatingly
fascination
fashionable
fashionably
```

```
bad
badly
baffle
baffled
bafflement
baffling
```

- Count **number of +/- words** in a review:
  - **Majority wins** (are there more + or - words?).
  - **Produce percentages** (x% positive, y% negative).

Lexicon-Based Sentiment Classification

+ No need for training data (reviews annotated as +/-), just lexicons.

+ Quick and easy to implement.

- Can't handle negations and other expressions, e.g.:

☺ The restaurant is <u>not</u> **bad** at all, so we will be back!

☺ I had heard that the service was **slow**, the food was **terrible**, and that it was **pricey**… but hey, it's actually **awesome**!

☹ **Fascinating** how this movie has such a **high rating** despite its plot.

Workaround for Negation

The restaurant is not bad at all, so we will be back!

Workaround: add “not_” to words after negation, up to next punctuation.

The restaurant is not **not_bad not_at not_all**, so we will be back!

↓

- **Expand lexicons to incorporate “not_*” in opposite list**,
  e.g.: *not_happy*, *not_good* as negative
  not_bad, *not_sad* as positive

Sentiment Classification – **Why is hard?**

- "If you are reading this because it is your darling fragrance, please wear it at home exclusively, and tape the windows shut."

  (review of the Givenchy perfume Amarige, in Perfumes: The Guide, Viking 2008.)

Sentiment Classification – **Why is hard?**

☹ "If you are reading this because it is your darling fragrance, please wear it at home exclusively, and tape the windows shut."

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- "She runs the gamut of emotions from A to B."

(Dorothy Parker, speaking about Katharine Hepburn.)

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- "Jane Austen's books madden me so that I can't conceal my frenzy from the reader. Every time I read 'Pride and Prejudice' I want to dig her up and beat her over the skull with her own shin-bone."

(Mark Twain)

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Lexicons ?!

fantastic
fantastically
fascinate
fascinating
fascinatingly
fascination
fashionable
fashionably

bad
badly
baffle
baffled
bafflement
baffling

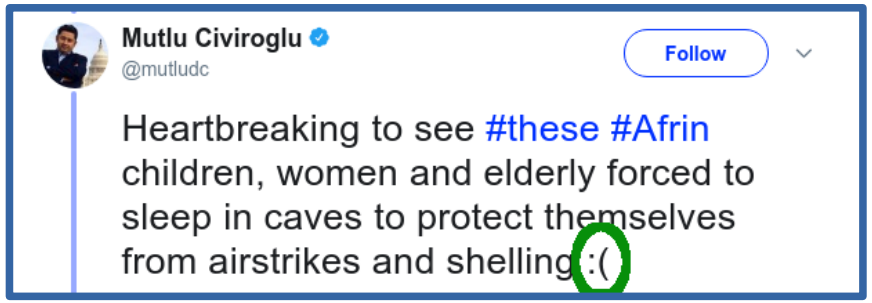
Learning-Based Sentiment Classification

- Pretty much like any text classification, we **need labelled data**.

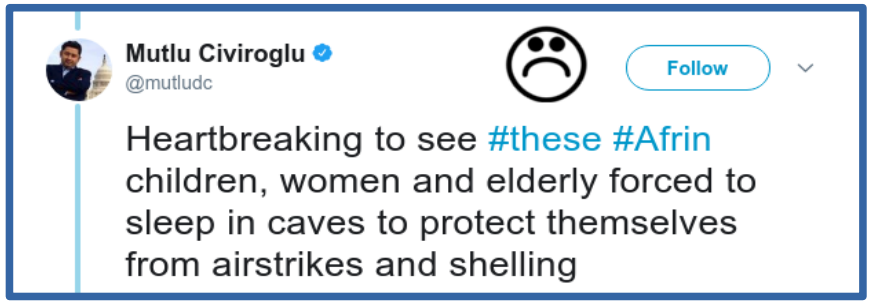
- Where we don’t have data, a good solution is **distant supervision**.

Distant Supervision

- Use positive and negative keywords to collect data, e.g.:
  `#happy` `:)` `#sad` `:(` …



- Build dataset after removing those keywords:



Distant Supervision

- + **Easy and quick** to get large collection of data.
- + (**Generally) free**, e.g. Twitter/X.
- - We can only get **positive/negative labels**, not neutral.
- - Data tends to be **noisy**, e.g. sarcasm.

Evaluation of Text Classification

Evaluation of Text Classification

Evaluation is different for binary and multiclass classification.

- **Binary**: we generally have a **positive** and a **negative class** (spam vs non-spam, medical test positive vs negative, exam pass vs fail).
  → Classification **errors can only go to <u>the other</u> class**.
- **Multiclass**: multiple categories, may have different level of importance.
  → Classification **errors can go to <u>any</u> other class**.

→ That is why we need to have a look at how to conduct proper evaluation.

Evaluation of Binary Classification

- 2-by-2 contingency table:

	Actually positive	**Actually negative**
Classified as positive	True Positive (TP)	False Positive (FP)
Classified as negative	False Negative (FN)	True Negative (TN)

Evaluation of Binary Classification

- 2-by-2 contingency table:

	Actually positive	**Actually negative**
Classified as positive	True Positive (TP)	Type I error (FP)
Classified as negative	Type II error (FN)	True Negative (TN)

Evaluation of Binary Classification

- 2-by-2 contingency table:

	Actually positive	Actually negative
Classified as positive	True Positive (TP)	False Positive (FP)
Classified as negative	False Negative (FN)	True Negative (TN)

- **Precision:** ratio of items classified as positive that are correct.

$$\text{Precision} = \frac{tp}{tp + fp}$$

- **Recall:** ratio of actual positive items that are classified as positive.

$$\text{Recall} = \frac{tp}{tp + fn}$$

Evaluation of Binary Classification

- We want to optimise for both precision and recall:

$$F = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$ (harmonic mean of precision and recall)

- Equation as follows, however generally ß = 1:

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}$$

Evaluation of Multiclass Classification

- Bigger **confusion matrix**:

	Actually UK	Actually World	Actually Tech	Actually Science	Actually Politics	Actually Business
Classified as UK	**95**	1	13	0	1	0
Classified as World	0	**1**	0	0	0	0
Classified as Tech	10	90	**0**	1	0	0
Classified as Science	0	0	0	**34**	3	7
Classified as Politics	0	1	2	13	**26**	5
Classified as Business	0	0	2	14	5	**10**

Overall Accuracy

- **Overall Accuracy:** ratio of correct classifications.

$$\frac{\sum_i c_{ii}}{\sum_j \sum_i c_{ij}}$$

Only when the dataset is **balance**!

- **Generally a bad evaluation approach**, e.g.:
  - Dataset with 1,000 emails: 970 are legitimate, 20 phishing, 10 spam.
  - Our classifier predicts all emails as legitimate.
  - 970 classified correctly: 970 / 1000 = 0.97 accuracy!

Misleading

Per-Class Precision / Recall

- **Per-class precision and recall**:

Precision: $\frac{c_{ii}}{\sum_j c_{ji}}$

c_{ii} → # of items correctly classified as class i

$\sum_j c_{ji}$ → # of items classified as class i

Recall: $\frac{c_{ii}}{\sum_j c_{ij}}$

c_{ii} → # of items correctly classified as class i

$\sum_j c_{ij}$ → # of actual class i items

- With the **harmonic mean**, we can then get **per-class F1 score**.

Overall Performance

- We have **per-class precision, recall** and **F1 scores**.
  - How do we combine them all to **get a single score**?

Overall Performance

Obtaining overall performances:

- **Macro-averaging:**
  Compute **performance independently for each class, then average them**.
  → All **classes are given equal weights** to the final score, regardless of their size (e.g. class with 990 and class with 10 instances).

- **Micro-averaging:**
  Compute **overall performance** without computing per-class performances.
  → **Large classes contribute more** to the final score.

Macro-Averaging

- Macro-averaging:

$$\text{macroaveraged precision} = \frac{\sum_i \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i}}{k}$$

$$\text{macroaveraged recall} = \frac{\sum_i \frac{\text{TP}}{\text{TP}_i + \text{FN}_i}}{k}$$

- The macro-averaged F1 score is then the harmonic mean of those.

Micro-Averaging

- Micro-averaging:

$$\text{microaveraged precision} = \frac{\sum_i \text{TP}_i}{\sum_i \text{TP}_i + \text{FP}_i}$$

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- The micro-averaged F1 score is then the harmonic mean of those.

Micro- vs. Macro-Averaging Example

	S	D	Q	C
S	**366 (40.4%)**	32 (3.5%)	22 (2.4%)	487 (53.7%)
D	38 (11.1%)	**22 (6.4%)**	23 (6.7%)	260 (75.8%)
Q	11 (3.1%)	10 (2.8%)	**149 (41.6%)**	188 (52.5%)
C	261 (9.0%)	91 (3.1%)	133 (4.6%)	**2,421 (83.3%)**

- Micro-averaged F1 score: **0.66**
- Macro-averaged F1 score: **0.45**

Q: where I have so many instances with label C, which metric is fairer?

A: depends on our objective: do we need a classifier that performs well for all labels? Do we just want to perform well for C?

Micro- vs. Macro-Averaging Example

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	S	Not S	Total
S	366	541	**907**
Not S	310	3297	**3607**
Total	**676**	**3838**	

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$$\text{Precision} = 366/907 = 0.40$$

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$$\text{Precision} = 366/907 = 0.40$$

$$\text{Recall} = 366/676 = 0.54$$

Macro-averaged F1 calculation

	S	Not S	Total
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Not S	310	3297	**3607**
Total	**676**	**3838**	

$$\text{Precision} = 366/907 = 0.40$$
$$\text{Recall} = 366/676 = 0.54$$

	D	Not D	Total
D	22	321	**343**
Not D	133	4038	**4171**
Total	**155**	**4359**	

$$\text{Precision} = 22/343 = 0.06$$
$$\text{Recall} = 22/155 = 0.14$$

	Q	Not Q	Total
Q	149	209	**358**
Not Q	178	3978	**4156**
Total	**327**	**4187**	

$$\text{Precision} = 149/358 = 0.42$$
$$\text{Recall} = 149/327 = 0.46$$

	C	Not C	Total
C	2421	485	**2906**
Not C	935	673	**1608**
Total	**3356**	**1158**	

$$\text{Precision} = 2421/2906 = 0.83$$
$$\text{Recall} = 2421/3356 = 0.72$$

$$\text{macroaveraged precision} = \frac{\sum_i \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i}}{k}$$

Macro-averaged F1 calculation

	S	Not S	Total
S	366	541	**907**
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Total	**676**	**3838**	

Precision = 366/907 = 0.40
Recall = 366/676 = 0.54

	D	Not D	Total
D	22	321	**343**
Not D	133	4038	**4171**
Total	**155**	**4359**	

Precision = 22/343 = 0.06
Recall = 22/155 = 0.14

	Q	Not Q	Total
Q	149	209	**358**
Not Q	178	3978	**4156**
Total	**327**	**4187**	

Precision = 149/358 = 0.42
Recall = 149/327 = 0.46

	C	Not C	Total
C	2421	485	**2906**
Not C	935	673	**1608**
Total	**3356**	**1158**	

Precision = 2421/2906 = 0.83
Recall = 2421/3356 = 0.72

MacroAvg – Precision

$$= \frac{0.40 + 0.06 + 0.42 + 0.83}{4}$$

$$= 0.43$$

$$\text{macroaveraged precision} = \frac{\sum_i \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i}}{k}$$

Macro-averaged F1 calculation

	S	Not S	Total
S	366	541	**907**
Not S	310	3297	**3607**
Total	**676**	**3838**	

$$\text{Precision} = 366/907 = 0.40$$
$$\text{Recall} = 366/676 = 0.54$$

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$$\text{Recall} = 2421/3356 = 0.72$$

$$\text{MacroAvg} - \text{Precision}$$
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$$= 0.43$$

Macro-averaged F1 calculation

$$\text{macroaveraged precision} = \frac{\sum_i \frac{\text{TP}_i}{\text{TP}_i+\text{FP}_i}}{k}$$

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MacroAvg – Recall

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$$\text{MacroAvg} - \text{Precision}$$

$$= \frac{0.40 + 0.06 + 0.42 + 0.83}{4}$$

$$= 0.43$$

$$\text{MacroAvg} - \text{Recall}$$

$$= \frac{0.54 + 0.14 + 0.46 + 0.72}{4}$$

$$= 0.47$$

$$\text{MacroAvg} - \text{F1}$$

$$= \frac{2 \times 0.43 \times 0.47}{0.43 + 0.47}$$

$$= 0.45$$

Macro-averaged F1 calculation

$$\text{macroaveraged precision} = \frac{\sum_i \frac{\text{TP}_i}{\text{TP}_i+\text{FP}_i}}{k}$$

$$\text{macroaveraged recall} = \frac{\sum_i \frac{\text{TP}}{\text{TP}_i+\text{FN}_i}}{k}$$

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Micro-averaged F1 calculation

$$\text{microaveraged precision} = \frac{\sum_i \text{TP}_i}{\sum_i \text{TP}_i + \text{FP}_i}$$

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Total	**3356**	**1158**	

Precision = 2421/2906 = 0.83
Recall = 2421/3356 = 0.72

MicroAvg – Precision

$$= \frac{366 + 22 + 149 + 2421}{907 + 343 + 358 + 2906} = 0.66$$

MicroAvg – Recall

$$= \frac{366 + 22 + 149 + 2421}{676 + 155 + 327 + 3356} = 0.66$$

MicroAvg – F1

$$= \frac{2 \times 0.66 \times 0.66}{0.66 + 0.66} = 0.66$$

Micro-averaged F1 calculation

$$\text{microaveraged precision} = \frac{\sum_i \text{TP}_i}{\sum_i \text{TP}_i + \text{FP}_i}$$

$$\text{microaveraged recall} = \frac{\sum_i \text{TP}}{\sum_i \text{TP}_i + \text{FN}_i}$$

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$$\text{MicroAvg} - \text{Precision}$$
$$= \frac{366 + 22 + 149 + 2421}{907 + 343 + 358 + 2906}$$
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$$\text{MicroAvg} - \text{Recall}$$
$$= \frac{366 + 22 + 149 + 2421}{676 + 155 + 327 + 3356}$$
$$= 0.66$$

$$\text{MicroAvg} - \text{F1}$$
$$= \frac{2 \times 0.66 \times 0.66}{0.66 + 0.66}$$
$$= 0.66$$

Further Weighting / Selection

- We can choose to **prioritise certain categories**:
  - Give higher weight to important categories:
    0.3*Prec(c_1) + **0.3***Prec(c_2) + **0.4***Prec(c_3)
    → unless we have clear criteria to choose these weights, it's rather arbitrary though!
  - Select some categories for inclusion in macro/microaverage:
    - e.g. Semeval task (Exercise 2 of the module), we only macroaverage over positive and negative sentiment classes. Performance over the neutral class is not included.

Error Analysis

Error Analysis

- **Error analysis:** can help us find out where our classifier can do better.
- No magic formula for performing error analysis.
  - Look where we are doing wrong, what labels particularly.
  - Do our errors have some common characteristics? Can we infer a new feature from that?
  - Could our classifier be favouring one of the classes (e.g. the majority class)?

Error Analysis

- **Error analysis:** where are we doing wrong? What labels?

  Look at frequent deviations in the confusion matrix.

	Actually UK	Actually World	Actually Tech	Actually Science	Actually Politics	Actually Business
Classified as UK	**95**	1	13	0	1	0
Classified as World	0	**1**	0	0	0	0
Classified as Tech	10	90	**0**	1	0	0
Classified as Science	0	0	0	**34**	3	7
Classified as Politics	0	1	2	13	**26**	5
Classified as Business	0	0	2	14	5	**10**

Error Analysis

Error analysis: could our classifier be favouring one of the classes?

- Owing to class imbalance, classifiers tend to predict ***popular classes*** more often.
  E.g.: class A (700), class B (100), class C (100), class D (100)
  classifiers will tend to predict A.

Over-represented as in our previous example:

	S	D	Q	C
S	**366 (40.4%)**	32 (3.5%)	22 (2.4%)	487 (53.7%)
D	38 (11.1%)	**22 (6.4%)**	23 (6.7%)	260 (75.8%)
Q	11 (3.1%)	10 (2.8%)	**149 (41.6%)**	188 (52.5%)
C	261 (9.0%)	91 (3.1%)	133 (4.6%)	**2,421 (83.3%)**

How to Deal with Class Imbalance?

How to deal with imbalance (e.g. **A-700, B-100, C-100, D-100**)?

- **Undersample popular class** → **A-100**, B-100, C-100, D-100

  randomly remove 600 instances of A

- **Oversample other classes** → A-700, **B-700, C-700, D-700**

  repeat instances of B, C, D to match the number of A's

- **Create synthetic data** → A-700, **B-700, C-700, D-700**

  generate new B, C, D items.

  Needs understanding of the contents of the classes to produce sensible data items

- **Cost sensitive learning**

  e.g. higher probability to predict uncommon classes

  P(A)=1/700, P(B)=1/100, P(C)=1/100, P(D)=1/100

  `scikit → class_weight="auto"`

How to Deal with Class Imbalance?

How to deal with imbalance (e.g. **A-700, B-100, C-100, D-100**)?

- **Undersample popular class** → **A-100**, B-100, C-100, D-100

  randomly remove 600 instances of A

- **Oversample other classes** → A-700, **B-700**, **C-700**, **D-700**

  repeat instances of B, C, D to match the number of A's

- **Create synthetic data** → A-700, **B-700, C-700, D-700**

  generate new B, C, D items.
  Needs understanding of the contents of the classes to produce sensible data items

- **Cost sensitive learning**

  e.g. higher probability to predict uncommon classes
  P(A)=1/700, P(B)=1/100, P(C)=1/100, P(D)=1/100
  `scikit → class_weight="auto"`

Error Analysis

- Important for the error analysis:
  - Subset we analyse for errors (dev set) has to be **different** to the one where we ultimately apply the classifier (test set).
  - If we tweak the classifier looking at the test set, we'll end up **overfitting**, developing a classifier that works very well for that particular test set.

- NB: for exercise 2 in the coursework, you're given 3 test sets, we'll test it in 2 more held-out test sets. Not looking at test sets while developing will improve generalisability.

Reading

- Jurafsky, Daniel and James H. Martin. 2026. Speech and Language Processing. 3rd edition. **Chapters 4-5**.

- Bird Steven, Ewan Klein, and Edward Loper. Natural Language Processing with Python. O'Reilly Media, Inc., 2009. **Chapter 6.**